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INTERDISCIPLINARY PROJECT WORK REPORT ON
“COLLEGE EVENT MANAGEMENT WEBSITE”

Submitted in partial fulfilment of the requirements for the award of the degree of
BACHELOR OF ENGINEERING
IN
DEPARTMENT OF BASIC SCIENCES
Submitted by

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CERTIFICATE

This is to certify that the IDP project entitled “College event management website” has been successfully carried out by MD DANISH HUSSAIN (1CR25A1066), MANYA M K (1CR25AI065), HARISANKAR L (1CR25IS066) and HARSHVARDHAN KUMAR (1CR25IS067) bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the IDP Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This IDP project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

EventHub.com is a centralized web-based platform developed to address the fragmented dissemination of inter-college event information across Bangalore's academic institutions. The project aims to simplify event discovery, registration, and participation for students by integrating technical, cultural, and sports events into a single digital ecosystem. The system employs modern web technologies, including Google OAuth authentication, RESTful API architecture, Cloudinary-based media management, QR-code-enabled attendance verification, and automated email notifications using Node mailer. A structured database schema was designed to manage users, events, and registrations efficiently, while organizer dashboards provide analytical insights into participant engagement and event performance. The platform also incorporates search and category filters to enhance accessibility and improve user experience. Testing and deployment were conducted across multiple devices to ensure reliability and responsiveness. The outcome of the project demonstrates improved visibility of inter-college opportunities, streamlined registration workflows, reduced duplication in participant records, and stronger collaboration between students and event organizers, thereby fostering a more connected and engaged campus community.

Keywords:

- Event management system,
- QR-based check-ins,
- Google OAuth,
- Student engagement,
- Centralized platform,
- Inter-college events

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Chapter 1

INTRODUCTION

College events such as technical fests, workshops, cultural programs, sports meets, and seminars play an important role in improving student skills, creativity, and teamwork. Managing these events manually using paper notices, spreadsheets, and physical registration forms often leads to communication gaps, data loss, registration errors, and delays in coordination. To overcome these challenges, a digital platform is required to simplify event planning and management. The College Event Management System (CEMS) is a web-based application designed to automate and centralize the entire event management process within an educational institution. The system allows organizers to create and manage events, students to register online, and administrators to monitor all event-related activities through a single platform. The proposed system improves communication, reduces manual effort, and provides real-time tracking of registrations and participation. It also supports secure authentication, automated notifications, QR-based ticket generation, and feedback collection after events.

Brief history of Event Management System

In earlier years, college events were managed using notice boards, manual registration books, and offline coordination methods. Event information was shared through printed posters and classroom announcements. As the number of events and participants increased, managing registrations and attendance became difficult.

With the growth of internet technologies and web applications, institutions started using digital tools for event registration and communication. Modern event management systems now provide online dashboards, automated ticketing, payment integration, and real-time analytics, making the entire process faster and more efficient.

Modern Event Management System

Modern event management systems use web and cloud technologies to simplify event organization. These systems allow users to access event information from any device and location. Features such as online registration, QR-code ticketing, payment gateways, notifications, and data analytics improve user experience and administrative efficiency. The College Event Management System uses technologies such as React.js, Node.js, and MongoDB/MySQL to provide a secure and user-friendly platform for managing college events.

Chapter 2

Problem Statement

2.1 Description

Managing college events through traditional methods creates several operational problems for organizers and students. Event announcements may not reach all students on time, registration details may be lost or duplicated, and manual attendance tracking consumes significant time. Handling payments, volunteer assignments, and participant coordination also becomes difficult when done manually. The lack of a centralized system reduces transparency and creates confusion regarding schedules, participation limits, and event updates. In addition, organizers often find it difficult to analyze participation data and feedback after the event.

2.2 Challenge Statement

The main challenge is to design and develop a centralized, secure, and efficient web-based platform that can automate event management activities such as event creation, participant registration, attendance tracking, notifications, and feedback collection while reducing manual effort and improving communication.

Chapter 3

3.1 Design Thinking Process

- a) Empathize: Discussions were conducted with students, faculty coordinators, and event organizers to understand the difficulties faced during event management. Common issues identified included delayed communication, registration errors, difficulty tracking attendance, and lack of centralized information.
- b) Define: The major requirements identified were:
 - Centralized event management platform
 - Easy online registration process
 - Real-time event updates
 - Secure login system
 - Attendance and participation tracking
 - Feedback and analytics support
- c) Ideate: Several ideas were considered, including mobile applications, spreadsheet automation, and cloud-based systems. The final solution selected was a web-based College Event Management System with role-based access and automated event handling features.
- d) Prototype: A prototype interface was developed with modules for administrators, organizers, and students. Features such as event dashboards, registration forms, QR ticket generation, and feedback forms were implemented.
- e) Test: The prototype was tested with sample users. Testing showed improved communication, faster registration, and easier event monitoring compared to manual methods.

3.2 Methodology

The working procedure of the College Event Management System is as follows:

1. Admin or organizer logs into the system.
2. Organizer creates and publishes a new event.
3. Students view available events on the dashboard.
4. Students register for selected events.
5. The system stores registration details in the database.
6. QR-based entry passes are generated automatically.
7. Organizers track registrations and attendance.
8. Participants provide feedback after the event.
9. Reports and analytics are generated for evaluation.

3.3 Prototype Description

The prototype consists of a web-based application developed using frontend and backend technologies connected to a database server.

3.3.1 Technologies Used

a) React.js

Used for developing the interactive frontend user interface.

b) HTML5 and CSS3

Used for webpage structure and styling.

c) Node.js with Express Framework

Used for backend development and API handling.

d) MongoDB / MySQL

Used for storing event details, user data, and registrations.

e) JWT Authentication

Provides secure user login and authentication.

f) Cloud Deployment Platform

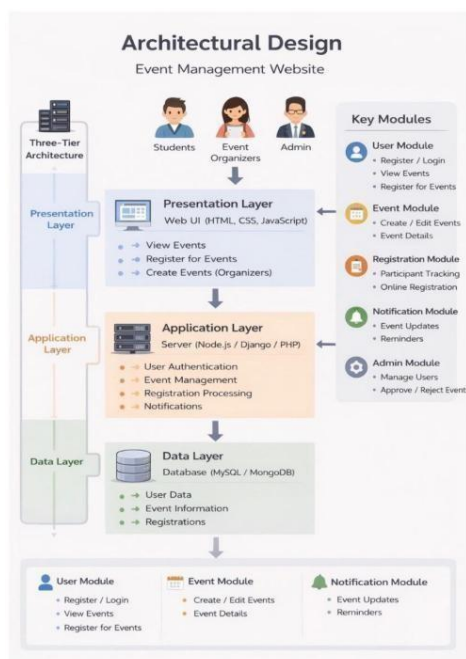
Platforms such as AWS, Vercel, or Heroku can be used for deployment.

3.3.2 System Diagram

The system architecture consists of three main layers:

1. Frontend Layer – Handles user interaction through web pages.
2. Backend Layer – Processes requests and business logic.
3. Database Layer – Stores user, event, and registration data.

The frontend communicates with the backend through APIs, and the backend interacts with the database for data storage and retrieval.



Chapter 4

Implementation

```
import type { Config } from "tailwindcss";

export default {

  content: [

    "./src/pages/**/*.{js,ts,jsx,tsx,mdx}",

    "./src/components/**/*.{js,ts,jsx,tsx,mdx}",

    "./src/app/**/*.{js,ts,jsx,tsx,mdx}",

  ],

  theme: {

    extend: {

      colors: {

        background: "hsl(var(--background))",

        foreground: "hsl(var(--foreground))",

        card: {

          DEFAULT: "hsl(var(--card))",

          foreground: "hsl(var(--card-foreground))",

        },

        popover: {

          DEFAULT: "hsl(var(--popover))",

          foreground: "hsl(var(--popover-foreground))",

        },

        primary: {
```

```
    DEFAULT: "hsl(var(--primary))",

    foreground: "hsl(var(--primary-foreground))",

  },

  secondary: {

    DEFAULT: "hsl(var(--secondary))",

    foreground: "hsl(var(--secondary-foreground))",

  },

  muted: {

    DEFAULT: "hsl(var(--muted))",

    foreground: "hsl(var(--muted-foreground))",

  },

  accent: {

    DEFAULT: "hsl(var(--accent))",

    foreground: "hsl(var(--accent-foreground))",

  },

  destructive: {

    DEFAULT: "hsl(var(--destructive))",

    foreground: "hsl(var(--destructive-foreground))",

  },

  border: "hsl(var(--border))",

  input: "hsl(var(--input))",

  ring: "hsl(var(--ring))",

  chart: {

    "1": "hsl(var(--chart-1))",
```

```
    "2": "hsl(var(--chart-2))",
    "3": "hsl(var(--chart-3))",
    "4": "hsl(var(--chart-4))",
    "5": "hsl(var(--chart-5))",
  },
},
borderRadius: {
  lg: "var(--radius)",
  md: "calc(var(--radius) - 2px)",
  sm: "calc(var(--radius) - 4px)",
},
},
},
plugins: [],
} satisfies Config;
```

Chapter 5

Results and Analysis

User Testing & Feedback

Sample Participants

- 15 students
- 3 faculty coordinators
- 2 event organizers

Quantitative Results

- Registration time reduced significantly.
- Event updates reached participants instantly.
- Manual paperwork was eliminated.
- Attendance tracking became faster using QR codes.
- Data accuracy improved compared to manual methods.

Qualitative Feedback

Students: “The online registration process is simple and convenient.”

Faculty Coordinators: “The dashboard makes monitoring events easier.”

Organizers: “Managing registrations and attendance has become more efficient.”

Software Output: Screenshots of the developed website such as login page, event dashboard, registration page, admin panel, and QR ticket generation can be included with explanations.

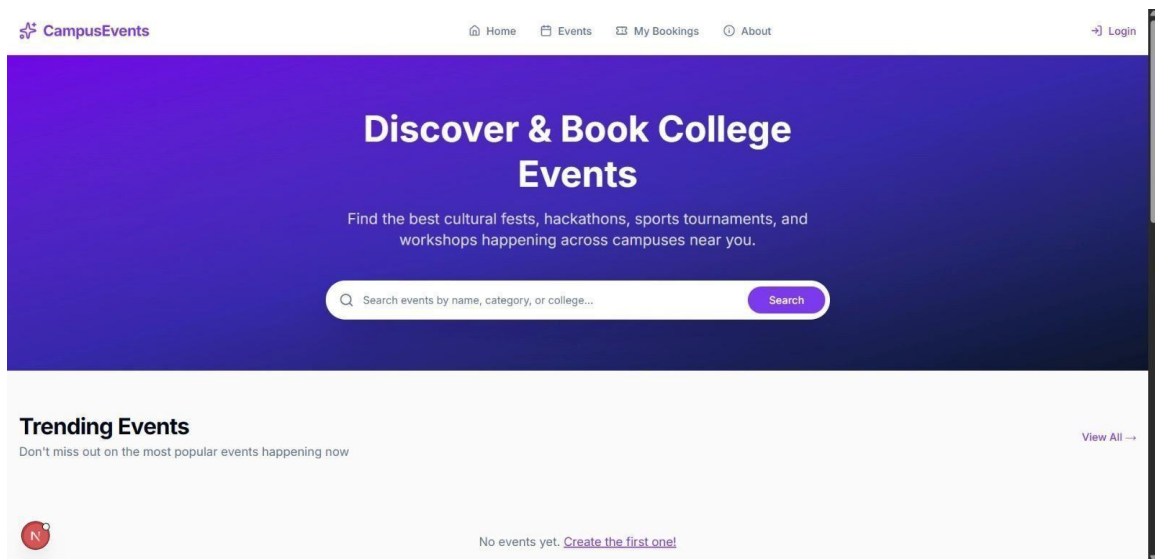
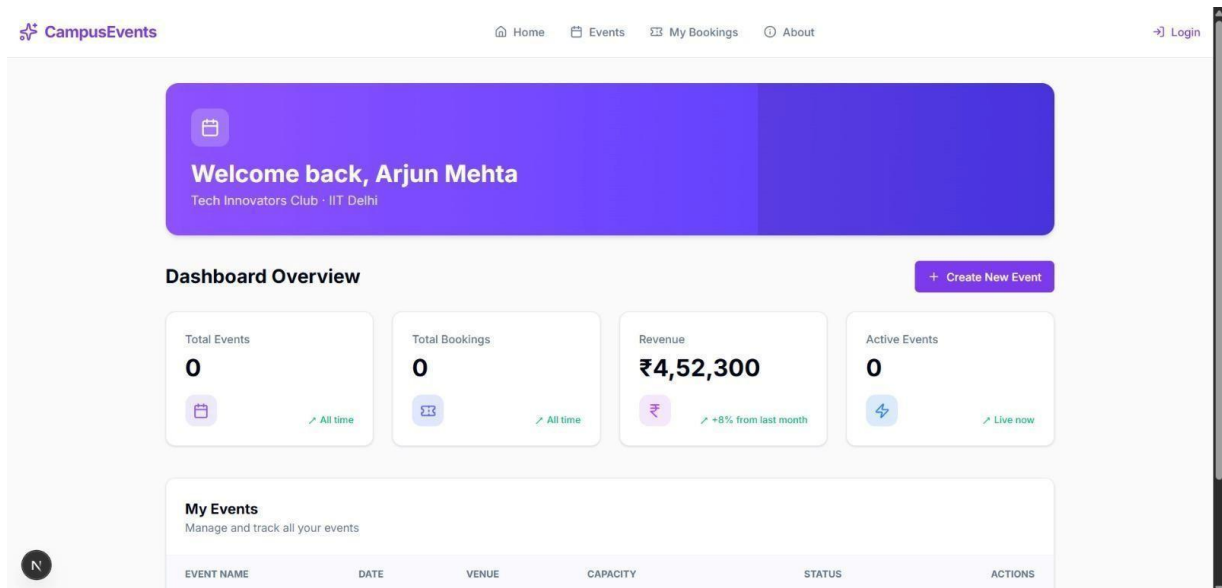
The results show that the College Event Management System successfully improves efficiency, communication, and event coordination within the institution.

Software Output

Screenshots of the developed website such as login page, event dashboard, registration page, admin panel, and QR ticket generation can be included with explanations. The results show that the College Event Management System successfully improves efficiency, communication, and event coordination within the institution.

The screenshot displays the 'Create New Event' interface of the CampusEvents website. The page features a navigation bar with links for Home, Events, My Bookings, About, and a Login button. The main heading is 'Create New Event', followed by the instruction 'Fill in the details to create and publish your college event'. A progress bar at the top indicates four steps: 1. Basic Info (active), 2. Details, 3. Tickets, and 4. Publish. The 'Basic Information' section includes a text input for 'Event Title' (with the example 'e.g., Annual Tech Fest 2025'), a dropdown menu for 'Event Category' (currently set to 'Cultural'), and a large dashed box for 'Event Banner Image' with a drag-and-drop icon and the text 'Drag & drop your image here'. To the right, a 'Preview' section shows a placeholder for the event title and date/venue, along with status indicators: 'No date set', 'No venue set', and 'No capacity defined'. A 'Tips for a Great Event' section at the bottom right provides advice: 'Use a clear, descriptive title' and 'Upload a 1920x1080 banner image'.

Project title: COLLEGE EVENT MANAGEMENT SYSTEM



Chapter 6

Conclusion & Future Work

The College Event Management System provides an effective solution for automating and managing college events. The system reduces manual work, improves communication between organizers and participants, and enables efficient registration and attendance tracking. By using web technologies and centralized data management, the platform enhances transparency and simplifies event coordination. The project successfully achieves its objectives of automation, accessibility, and real-time monitoring. The developed system can be used by educational institutions to manage technical, cultural, and sports events more effectively.

Future Work

- Development of Android and iOS mobile applications
- AI-based personalized event recommendations
- Integration with live streaming platforms
- Advanced analytics and reporting features
- Integration with college ERP systems
- Push notification support for instant updates

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Research Papers

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- Raghu Ramakrishnan and Johannes Gehrke: Database Management Systems, 3rd Edition, McGraw Hill, 2003.
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1. K.R. Venugopal, K.G. Srinivas, P.M. Krishnaraj: Database Management Systems.
2. Web Technologies by Uttam K. Roy.

Online Resources

1. <https://react.dev/>
2. <https://nodejs.org/>
3. <https://expressjs.com/>
4. <https://www.mongodb.com/>
5. <https://developer.mozilla.org/>

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles